

Section 6.1 — Discrete Random Variables: Every Question Solved Step-by-Step

Covers HW 6.1.1, 6.1.3, 6.1.5, 6.1.19, 6.1.21, 6.1.25, 6.1.27, 6.1.30 • Read THE 4 FACTS box first — every answer below comes from it. Blue boxes = exact StatCrunch clicks (no typing expressions). Yellow boxes = traps that cost points. Δ MyLab shuffles answer letters between versions — always match by the wording/graph described, never by the letter.

THE 4 FACTS THAT SOLVE EVERYTHING

1. Random variable (RV): a numerical measure of the outcome of a probability experiment — a number whose value is determined by chance.

2. Discrete vs. continuous: ask "Do I COUNT it or MEASURE it?" Counted with gaps (0, 1, 2, ...) → **discrete**. Measured on a scale where any decimal is possible (rain, time, weight) → **continuous**.

3. Legal probability distribution — 2 rules: every probability is between 0 and 1 inclusive ($0 \leq P(x) \leq 1$), and all probabilities add to exactly 1 ($\Sigma P(x) = 1$).

4. Formulas:

$$\mu_X = \Sigma[x \cdot P(x)] \quad \sigma_X = \sqrt{(\Sigma[x^2 \cdot P(x)] - \mu_X^2)}$$

Mean = "multiply each value by its probability, add them up." You will let StatCrunch do both (recipe below) — but the multiply-and-add idea is what the mean IS.

MASTER STATCRUNCH RECIPE (use on Q4–Q8)

1. Type the x-values in **var1** and the probabilities P(x) in **var2** (one row per outcome).

2. Stat → Calculators → Custom. Set **Values in: var1, Weights in: var2** → **Compute!**

3. The top of the window prints **Mean** ($= \mu_X$) and **Std. Dev.** ($= \sigma_X$). Done — no formulas typed.

4. For probabilities: in the same window use the **P(X ? x)** box — pick =, <, ≤, >, or ≥, type the value, Enter. It shades the bars and prints the answer.

Frequencies instead of probabilities? The Weights column also accepts *raw frequencies* — StatCrunch rescales them automatically (see Q5). Money problems? Build the outcome table first (see Q6).

Q1 · 6.1.1 — "What is a random variable?" definition

A — a numerical measure of the outcome of a probability experiment.

Why: that's the definition (Fact 1), word for word. Its value is determined by chance.

Trap: B adds "plotted on a line in an uninterrupted fashion" → that defines a *continuous* RV only. C adds "a finite or countable number of values" → that defines a *discrete* RV only. The general definition has no such restriction — pick the plain one.

Q2 · 6.1.3 — Two requirements for a discrete probability distribution select all

$\Sigma P(x) = 1$ and **$0 \leq P(x) \leq 1$** (choices C and D)

Why: exactly Fact 3: probabilities each live between 0 and 1 inclusive, and together they must account for 100% of outcomes (sum = 1).

Trap: " $\Sigma P(x) = 0$ " is nonsense (total probability is 1, not 0). " $0 < P(x) < 1$ " is wrong because it's *strict*: a probability is allowed to equal 0 or 1, so the ≤ version is the correct rule.

Q3 · 6.1.5 — Discrete or continuous? classify + list values

(a) Number of people in a restaurant (capacity 100): you count people → **discrete; x = 0, 1, 2, ..., 100** (choice A)

Counts start at 0 and stop at the capacity 100; only whole numbers (no 2.5 people).

(b) Inches of rain in City B during April: you measure rain — 2.37 in. is possible → **continuous; r ≥ 0** (choice B)

Any value in an interval; can't be negative; no fixed upper limit.

Trap: in these answer choices, interval notation like " $0 \leq x \leq 100$ " (every real number between 0 and 100) is reserved for continuous variables — a discrete answer is written as the list 0, 1, 2, ..., 100. Match the notation style to your discrete/continuous choice.

Q4 · 6.1.19 — Baseball hits distribution the full workout: verify, graph, μ , σ , probabilities

x (hits)	P(x)	x · P(x)
0	0.1675	0.0000
1	0.3357	0.3357
2	0.2851	0.5702
3	0.1482	0.4446
4	0.0384	0.1536
5	0.0251	0.1255
Sum	1.0000	1.6296

(a) Verify it's a probability distribution: check the 2 rules (Fact 3). Every P(x) here is between 0 and 1 ✓; add the P(x) column:

$0.1675 + 0.3357 + 0.2851 + 0.1482 + 0.0384 + 0.0251 = 1.0000$ ✓. **Fill-in: probabilities are between 0 and 1, inclusive, and the sum of the probabilities is 1.**

(b) Pick the correct graph: the bar at each x must have height exactly P(x). Quick method: find the **largest P** — here 0.3357 at x = 1 — so the tallest bar must sit at x = 1, then heights step down: 0.2851, 0.1482, ... **Graph C.** Shape: single peak (x = 1) = **one mode**; the long thin tail runs to the *right* (high x, tiny probabilities) → **skewed right**.

(c) Mean: multiply each x by its P(x) (3rd column) and add: **$\mu_X = 1.6296$ hits** (do NOT round — it said so). **Interpret** = choice **D**: "over the course of many games, one would expect the mean number of hits per game to be the mean of the random variable."

Trap: interpretation answers with "most games", "will equal", or "less than" are wrong. The mean is a **long-run average** (Law of Large Numbers): correct wording always = "many repetitions" + "average ≈ 1.6296." A player can't get 1.6296 hits in one game.

(d) Standard deviation: StatCrunch recipe → Std. Dev. = 1.1816... → **$\sigma_X \approx 1.182$ hits** (round 3 decimals as instructed).

(e) P(exactly 2 hits): "exactly" = read one row of the table: **P(2) = 0.2851**.

(f) P(more than 1 hit): "more than 1" means x = 2, 3, 4, 5 (NOT 1). Add: $0.2851 + 0.1482 + 0.0384 + 0.0251 = 0.4968$. Shortcut: $1 - P(0) - P(1) = 1 - 0.1675 - 0.3357 = 0.4968$. ✓

StatCrunch: x in var1, P(x) in var2 → Stat → Calculators → Custom → Values var1, Weights var2 → Compute. Read Mean = 1.6296, Std. Dev. = 1.182. For (e): P(X = 2); for (f): P(X > 1) using the dropdown. Zero hand arithmetic.

Q5 · 6.1.21 — Games per series, 1923–2022 frequencies → distribution → μ , σ

Setup: total series $N = 18+23+25+33 = 99$.

(a) Build $P(x)$ = frequency ÷ N (round to 4 decimals):

x (games)	freq	$P(x) = \text{freq}/99$
4	18	$18/99 = 0.1818$
5	23	$23/99 = 0.2323$
6	25	$25/99 = 0.2525$
7	33	$33/99 = 0.3333$

(b) Graph: heights must climb to the tallest bar at $x = 7$ ($P = 0.3333$): $0.18 \rightarrow 0.23 \rightarrow 0.25 \rightarrow 0.33$. **Graph D.**

(c) Mean: $\mu = 4(0.1818) + 5(0.2323) + 6(0.2525) + 7(0.3333) \approx 5.74 \rightarrow \mu_x \approx 5.7$ games (1 decimal). **Interpret** = choice **A**: "the series, if played many times, would be expected to last about 5.7 games, on average."

(d) Standard deviation: StatCrunch → $\sigma_x \approx 1.1$ games (1 decimal).

StatCrunch: put 4,5,6,7 in var1 and the raw frequencies 18,23,25,33 in var2 → Custom calculator → Mean 5.7, Std. Dev. 1.1 (it rescales weights automatically — no dividing needed for μ and σ). For part (a)'s table you still divide: do $18 \div 99$ on a plain calculator and type the decimal **0.1818** into the answer box. Never type "18/99" into a StatCrunch cell — cells only accept plain numbers.

Q6 · 6.1.25 — Insurance expected value \$290,000 policy, \$360 premium, $P(\text{survive}) = 0.999544$

Step 1 — outcomes from the COMPANY's point of view (this is the whole trick):

Event	Company's profit x	$P(x)$
She survives	+\$360 (keeps premium)	0.999544
She dies	$360 - 290,000 = -\$289,640$	$1 - 0.999544 = 0.000456$

Step 2 — expected value $= \sum x \cdot P(x) = 360(0.999544) + (-289,640)(0.000456) = 359.84 - 132.08 = \227.76 . Shortcut: $E = 360 - 290,000(0.000456) = 360 - 132.24 = 227.76$ — the shortcut uses the FULL payout 290,000 because the company keeps the \$360 premium either way; don't mix it with the table's net -289,640.

Step 3 — interpret = choice **C**: "the insurance company expects to make a profit of \$227.76 on every 20-year-old female it insures for 1 year."

Trap: "maximum profit" / "minimum profit" / "1 month" are all wrong — expected value is an **average per policy per year**, not a max or a guarantee.

StatCrunch: var1 = 360 and -289640; var2 = 0.999544 and 0.000456 → Custom calculator → Mean = 227.76.

Q7 · 6.1.27 — Insurance standard deviation \$240,000 policy, \$190 premium, $E = \$80.56$ given

Step 1 — same two-row table: survives → +\$190 with 0.999544; dies → $190 - 240,000 = -\$239,810$ with 0.000456.

Step 2 — σ by StatCrunch (never by hand — the numbers are huge): Custom calculator → Std. Dev. = 5123.8 → $\sigma \approx \$5124$ (nearest dollar).

Step 3 — why so high? = choice **C**: "because the value of the payout is much larger than the value of the gain." σ measures spread — the two outcomes (+190 vs -239,810) are ~\$240,000 apart, so the spread is enormous even though a payout is very unlikely.

Trap: "probability of a payout is so low/high" explains why the mean is a small profit — it does NOT explain the spread. Distance between outcomes drives σ .

Q8 · 6.1.30 — Expected profit of an investment 3 outcomes

Step 1 — outcome table (losses are NEGATIVE x-values):

Economy	x (profit)	$P(x)$	$x \cdot P(x)$
Strong	+30,000	0.20	+6,000
Moderate	+20,000	0.60	+12,000
Recession	-30,000	0.20	-6,000

Step 2 — add the last column: $6,000 + 12,000 - 6,000 = E = \$12,000$.

Sanity checks: probabilities sum to $0.20+0.60+0.20 = 1$ ✓; answer must land between -30,000 and +30,000 ✓.

StatCrunch: var1 = 30000, 20000, -30000; var2 = 0.2, 0.6, 0.2 → Custom calculator → Mean = 12000.

EXAM PLAYBOOK — how to replicate this on ANY 6.1 problem

- See a table of x and $P(x)$?** → StatCrunch Custom calculator immediately: gives μ , σ , and any $P(X...)$ with zero formulas. "Exactly a" = one row; "more than a" excludes a; "at least a" includes a.
- Verify a distribution?** → recite the 2 rules: each $0 \leq P(x) \leq 1$, and $\sum P(x) = 1$ (add them up and show it). If YOU rounded the $P(x)$ s, a sum of 0.9999 or 1.0001 still counts as 1 — that's rounding error, not an invalid distribution.
- Given frequencies?** → $P(x) = \text{freq} \div \text{total}$, round as told, then proceed as above.
- Pick a graph?** → match the tallest bar to the biggest $P(x)$, then check each height. Peak near left with long right tail = skewed right; mirror = skewed left; ~symmetric = symmetric. Modes = number of peaks.
- Money/insurance?** → 2-row profit table from the point of view the question names (these HW ask the COMPANY's): [+premium, $P(\text{survive})$] and [premium - payout, $1 - P(\text{survive})$]. If asked from the BUYER's side, signs flip. Investments: list each profit (loss = negative) with its probability. Then Custom calculator.
- Interpretation MCQ?** → the right answer always says "over MANY repetitions the AVERAGE is about μ ." Kill any choice saying most / exactly / guaranteed / one trial / maximum / minimum / wrong time unit.
- Rounding:** obey the blue on-screen instruction exactly ("do not round" vs "3 decimals" vs "nearest cent/dollar").